

# LESSON 2-4 • SESSION 2 • PRACTICE • FORM A

## Comparing Box Plots

Learning target – I can compare two groups by reading the medians and box widths of their box plots. · 6.SP.B.4

### REMEMBER

- Put both box plots on the **same scale**.
- Compare **medians** for center: which group is typically higher, and by how much?
- Compare **box widths** ( $Q3 - Q1$ ) for spread: the narrower box is more **consistent**.

### WORKED EXAMPLE

Trail X: Q1 18, median 20, Q3 22. Trail Y: Q1 19, median 26, Q3 33 (minutes).

**Step 1** Medians:  $26 - 20 = 6$ , so Y is typically 6 min longer.

**Step 2** Box widths: X is  $22 - 18 = 4$ ; Y is  $33 - 19 = 14$ .

**Answer** Y is longer; X is more consistent

**WORDS** center (*centro*) · spread (*dispersión*) · consistent (*constante*)

### PRACTICE

show your work · label units

- 1. Two Quizzes.** The box plots show scores on two quizzes.

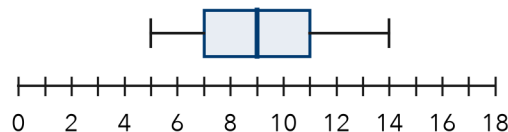
**Hint** The line inside each box is the median.

- a. Higher median: \_\_\_\_\_
- b. By how many points: \_\_\_\_\_
- c. Wider box: \_\_\_\_\_

Quiz 2



Quiz 1



Score

- 2. Two Teams.** Team Red points: 42, 50, 46, 54, 48, 56, 52. Team Blue points: 41, 45, 49, 43, 55, 37, 58. Complete the table. Which team was more consistent?

**Hint** The narrower box is the more consistent team.

| Team | Min | Q1 | Median | Q3 | Max |
|------|-----|----|--------|----|-----|
| Red  |     |    |        |    |     |
| Blue |     |    |        |    |     |

**More consistent** \_\_\_\_\_

- 3. Same Median.** Two plots have a median of 30. Box C runs from 27 to 33. Box D runs from 18 to 42.

- a. Width of box C: \_\_\_\_
- b. Width of box D: \_\_\_\_

- 4. More Consistent?** Plot P has  $Q1 = 22$  and  $Q3 = 30$ . Plot Q has  $Q1 = 18$  and  $Q3 = 36$ . Which middle half is more consistent?

**Hint** Find each box width:  $Q3 - Q1$ .

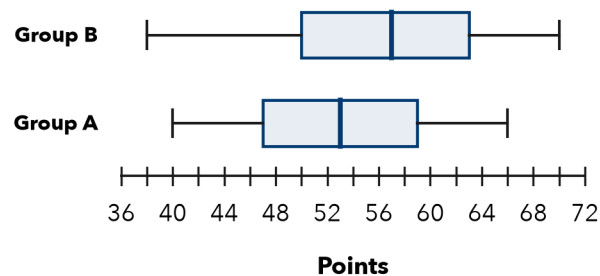
☐ P ☐ Q

**Widths** \_\_\_\_\_

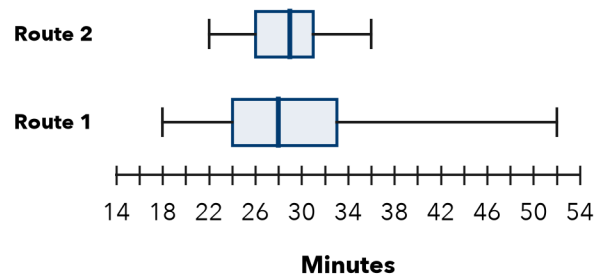
- 5. Equal Spread?** Plot M runs from 20 to 90 with a box from 45 to 65. Plot N runs from 50 to 70 with a box from 55 to 60. Jamal says they are equally spread. Is he right? Compare the spans and the boxes.

☐ Yes, equal spread ☐ No, M is more spread out ☐ No, N is more spread out

6. **Two Groups.** Which group has the higher median, and by how much?  
 Group; difference \_\_\_\_\_



7. **Matching Boxes.** Plot W runs from 10 to 60. Plot Z runs from 28 to 44. Both boxes run from 30 to 40. Omar says they show the same spread.
- a. Whole span of W (max – min): \_\_\_\_\_
  - b. Whole span of Z: \_\_\_\_\_
  - c. Is Omar right? Explain.
8. **Two Bus Routes.** Travel times (minutes) for two routes.
- a. Higher median: \_\_\_\_\_
  - b. More predictable (use box widths): \_\_\_\_\_
  - c. A rider must not be late. Which route? Why?



**FIND THE ERROR** explain the mistake, then fix it

9. **Bianca's Work.** Comparing two box plots, Bianca wrote: "Plot E has a longer right whisker, so Plot E has the greater median." Explain the mistake.
- \_\_\_\_\_
- \_\_\_\_\_

**TEST PRACTICE** choose the best answer

10. The line inside a box plot's box shows the –
- A** mean                      **B** median                      **C** range                      **D** Q1
11. Group A's median is 50 and Group B's is 62. Group B's center is higher by –
- A** 12                      **B** 112                      **C** 62                      **D** 10

**SELF-CHECK CODE** find your answer · copy its letter into that problem's box · spell a math word

|                       |                            |                            |                      |                       |                        |                      |                         |                       |                       |
|-----------------------|----------------------------|----------------------------|----------------------|-----------------------|------------------------|----------------------|-------------------------|-----------------------|-----------------------|
| <b>24</b><br><b>R</b> | <b>Route 1</b><br><b>X</b> | <b>Route 2</b><br><b>S</b> | <b>1</b><br><b>S</b> | <b>60</b><br><b>Q</b> | <b>Red</b><br><b>P</b> | <b>4</b><br><b>E</b> | <b>Blue</b><br><b>Z</b> | <b>50</b><br><b>A</b> | <b>16</b><br><b>D</b> |
| #1b                   |                            | #2                         | #3b                  | #6                    | #7a                    | #7b                  | #8b                     |                       |                       |

**CHECK YOURSELF**    ☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about # \_\_\_\_\_